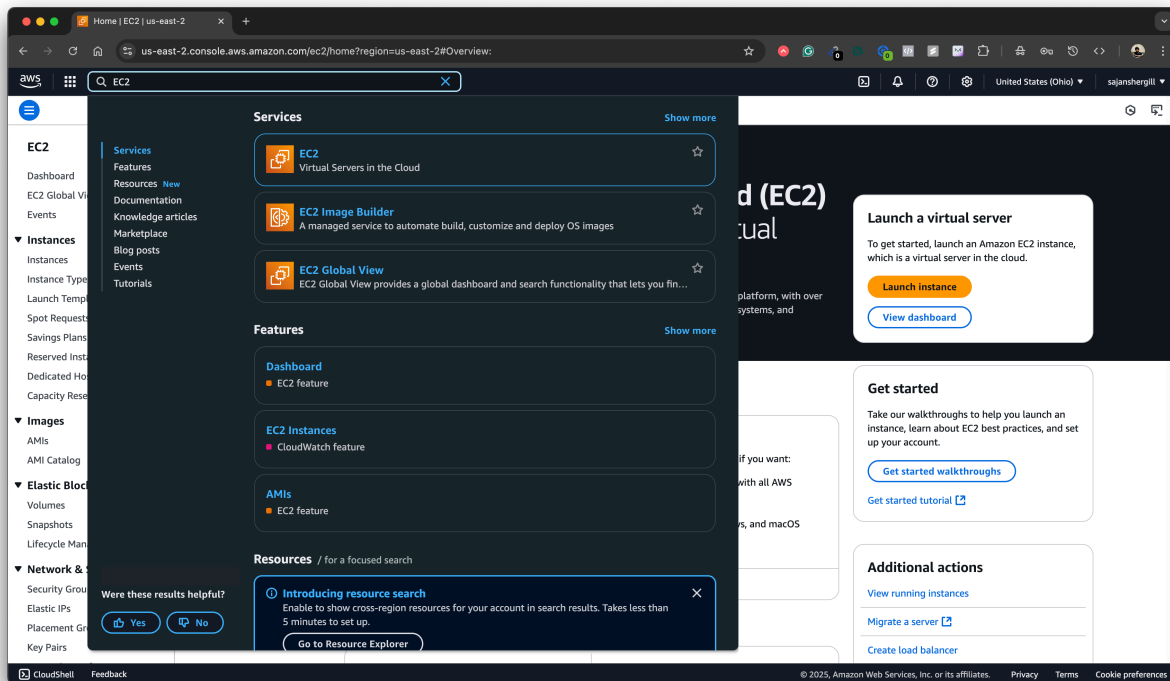


Creation EC2 Key Pair

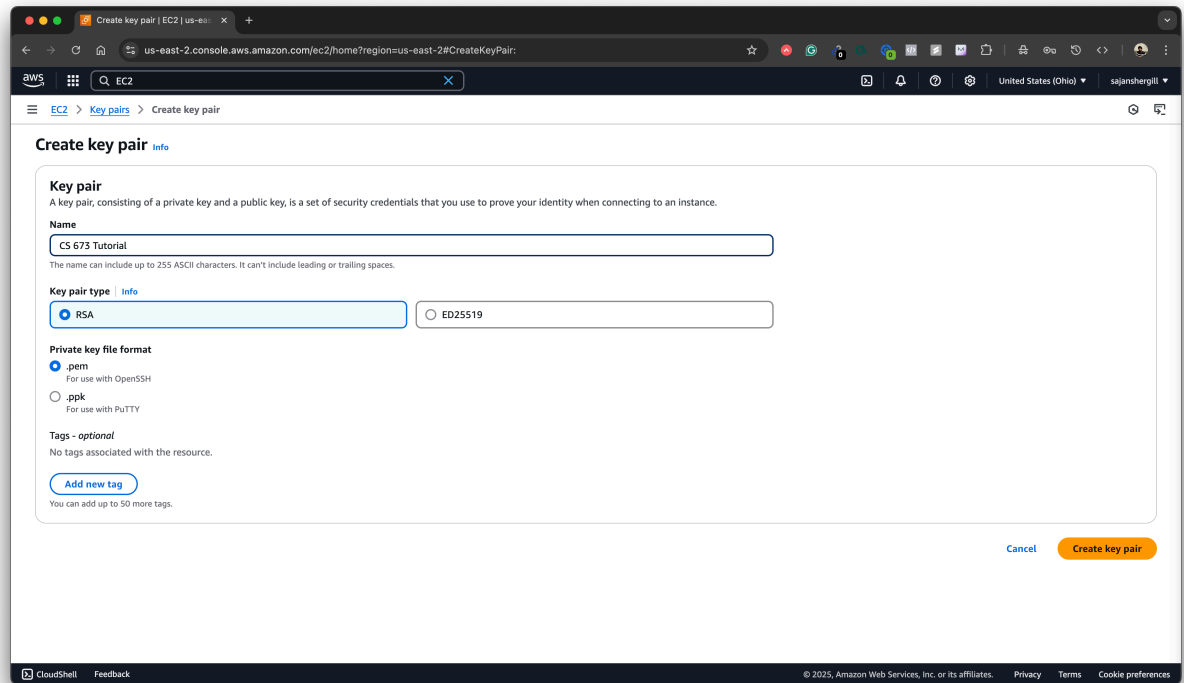
1. Creation an “EC2 Key Pair” to connect to the EMR cluster via any SSH client
 - a. Open the Amazon console and type “EC2” in the search bar of the dashboard, select it.
 - b. In the navigation pane, under Network & Security, choose Key Pairs



- c. Choose Create key pair.

For Name, enter a descriptive name for the key pair. Amazon EC2 associates the public key with the name that you specify as the key name.

“pem” configuration is more suitable for Mac/Linux

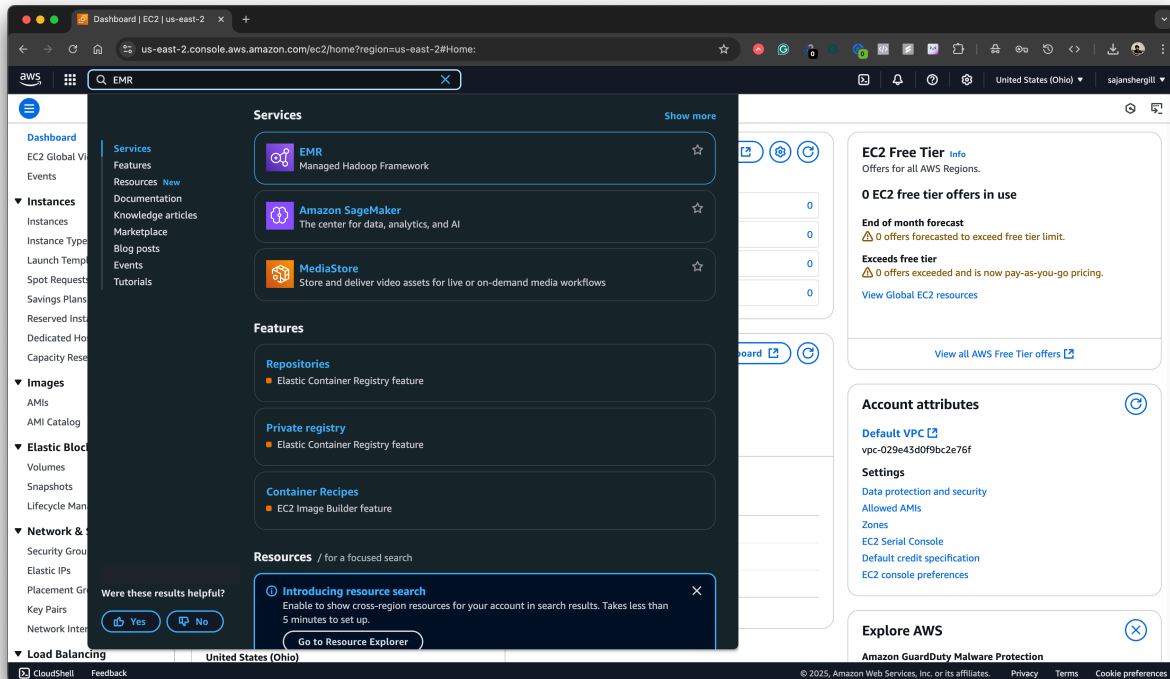


d. For Key pair type, choose either RSA

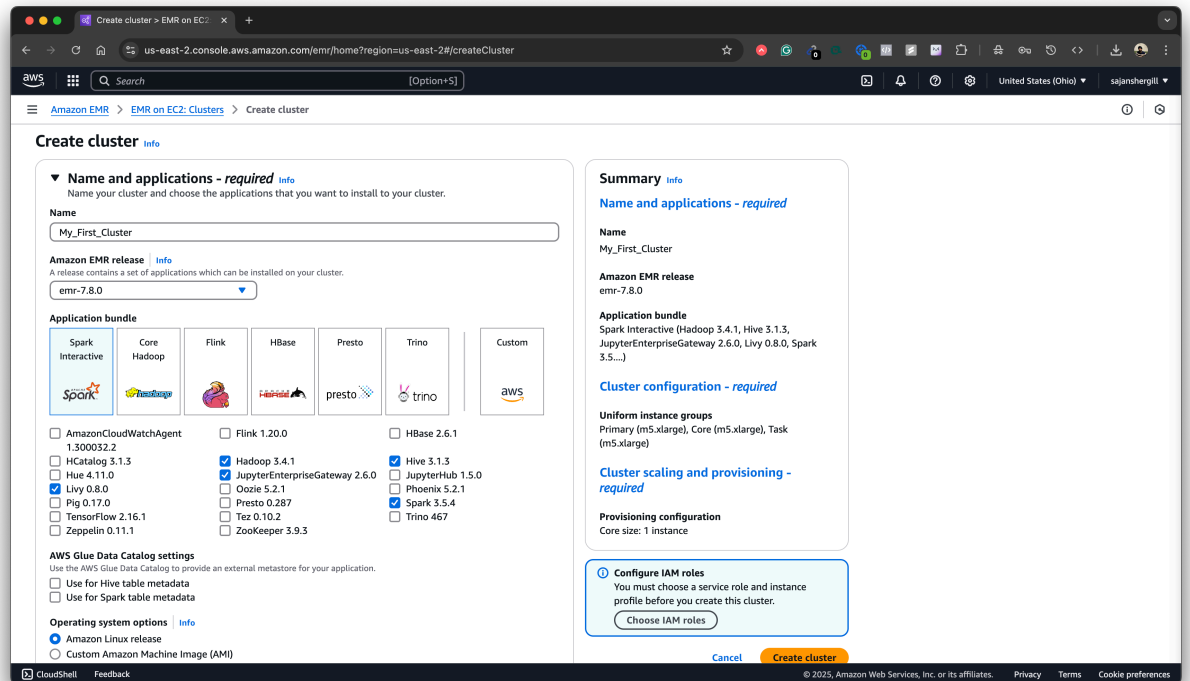
e. The private key file is automatically downloaded by your browser. The base file name is the name that you specified as the name of your key pair, and the file name extension is determined by the file format that you chose. Save the private key file in a safe place.

Creation the EMR cluster

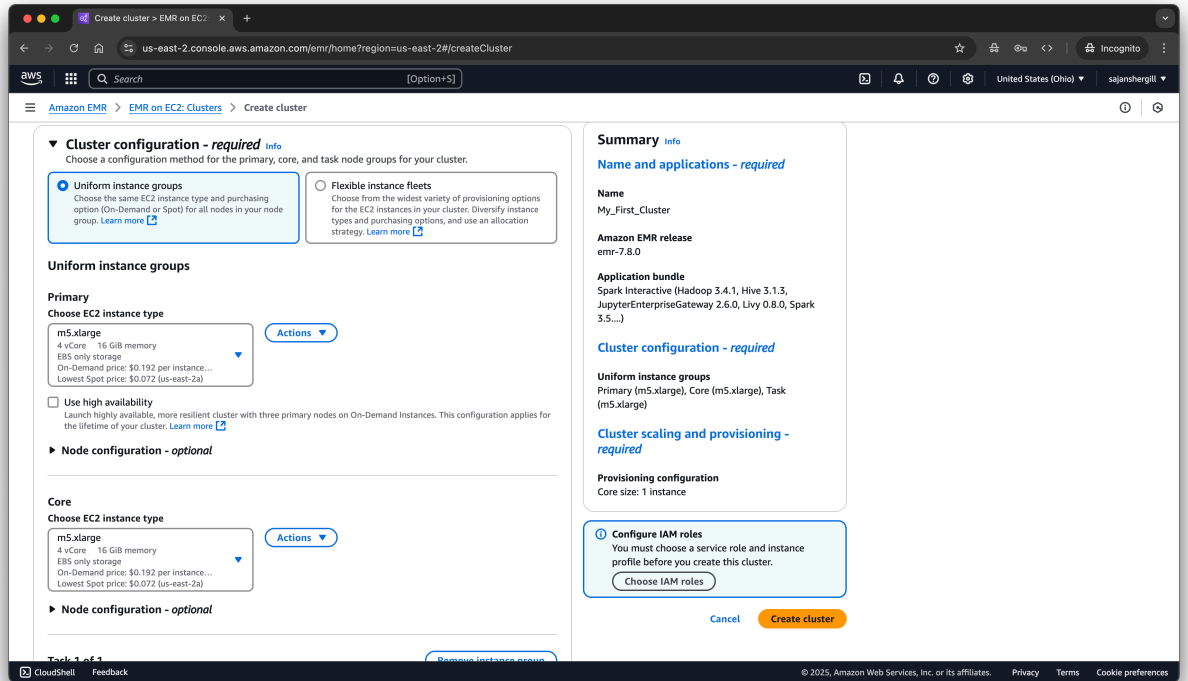
2. Creation the EMR cluster. First of all, please change your region to “Ohio”
 - a. Open the Amazon console and type “EMR” in the search bar of the dashboard, select it.



- b. Choose Create cluster.
Provide a Name to the cluster and select any Release of EMR (Ex: emr-7.0.0)
Keep the instance type as m5.xlarge and you can change the “Number of Instances” as per your requirement. It will be sufficient 1 instance for us.



Keep the instance type as m5.xlarge and you can change the “Number of Instances” as per your requirement. It will be sufficient 1 instance for us. Keep remained other settings that were installed by default.



Within “EC2 key pair” field; select the key pair which you created in Step 1.

▼ Security configuration and EC2 key pair [Info](#)

Choose a security configuration or create a new one that you can reuse with other clusters.

Security configuration

Select your cluster encryption, authentication, authorization, and instance metadata service settings.

[Refresh](#) [Browse](#) [Create security configuration](#)

Amazon EC2 key pair for SSH to the cluster [Info](#)

[X](#) [Browse](#) [Create key pair](#)

Select Service role - “EMR_DefaultRole”

Select Instance profile - “DemoRoleForEC2”

If you do not have “Default and Demo Roles”, please click on “Create a service role” and on “Create an instance profile” by keeping all default configurations.

After clicking on Create Cluster, it will take around 10- 15 minutes for the cluster to go in running state. In the meantime, follow step 3

▼
Identity and Access Management (IAM) roles - required
Info

Choose or create a service role and instance profile for the EC2 instances in your cluster.

Amazon EMR service role
Info

The service role is an IAM role that Amazon EMR assumes to provision resources and perform service-level actions with other AWS services.

☐
Choose an existing service role

Select a default service role or a custom role with IAM policies attached so that your cluster can interact with other AWS services.

☒
Create a service role

Let Amazon EMR create a new service role so that you can grant and restrict access to resources in other AWS services.

Networking resources

We've already added the resources that you configured in the Networking section. Choose the VPC, subnet, and security groups that the service role can access.

Virtual Private Cloud (VPC)

Choose one or more VPCs

-
vpc-029e43d0f9bc2e76f

Subnet

Choose one or more subnets

-
subnet-001e8847fd20b23be

Security group

Choose one or more security groups

default
sg-02aaa3be3021b3cfe

Cluster termination

Cluster termination

Terminate cluster after idle time

Idle time: 1 hour

Cluster logs

Amazon S3 location

s3://aws-logs...

Security configuration and EC2 key pair

Amazon EC2 key pair

CS 673 Tutori...

Identity and Access Management (IAM) roles - required

Service role

New service role

Instance profile

New instance profile

Cancel

Create cluster

After clicking on Create Cluster, it will take around 10- 15 minutes for the cluster to go in running state. In the meantime, follow step 3

39-029e43d0f9bc2e76f

EC2 instance profile for Amazon EMR

The instance profile assigns a role to every EC2 instance in a cluster. The instance profile must specify a role that can access the resources for your steps and bootstrap actions.

☐
Choose an existing instance profile

Select a default role or a custom instance profile with IAM policies attached so that your cluster can interact with your resources in Amazon S3.

☒
Create an instance profile

Let Amazon EMR create a new instance profile so that you can specify a custom set of resources for it to access in Amazon S3.

S3 bucket access
Info

☒
Specific S3 buckets or prefixes in your account
Info

Choose the buckets or prefixes that you want this instance profile to access.

☐
All S3 buckets in this account with read and write access

Grant the instance profile access to all buckets that have read and write access enabled in your account.

S3 buckets

We've already added the resources that you configured in the Cluster logs section. Choose the S3 buckets and bucket prefixes where you store logs and data for your cluster, bootstrap actions, and steps.

S3 URI

s3://bucket/prefix/object

View

Browse S3

Add

S3 bucket	Prefix	Permission	
aws-logs-41148140...	elasticmapreduce	Read and write	Edit
Inherited from Cluster logs			

Cluster termination

Cluster termination

Terminate cluster after idle time

Idle time: 1 hour

Cluster logs

Amazon S3 location

s3://aws-logs...

Security configuration and EC2 key pair

Amazon EC2 key pair

CS 673 Tutori...

Identity and Access Management (IAM) roles - required

Service role

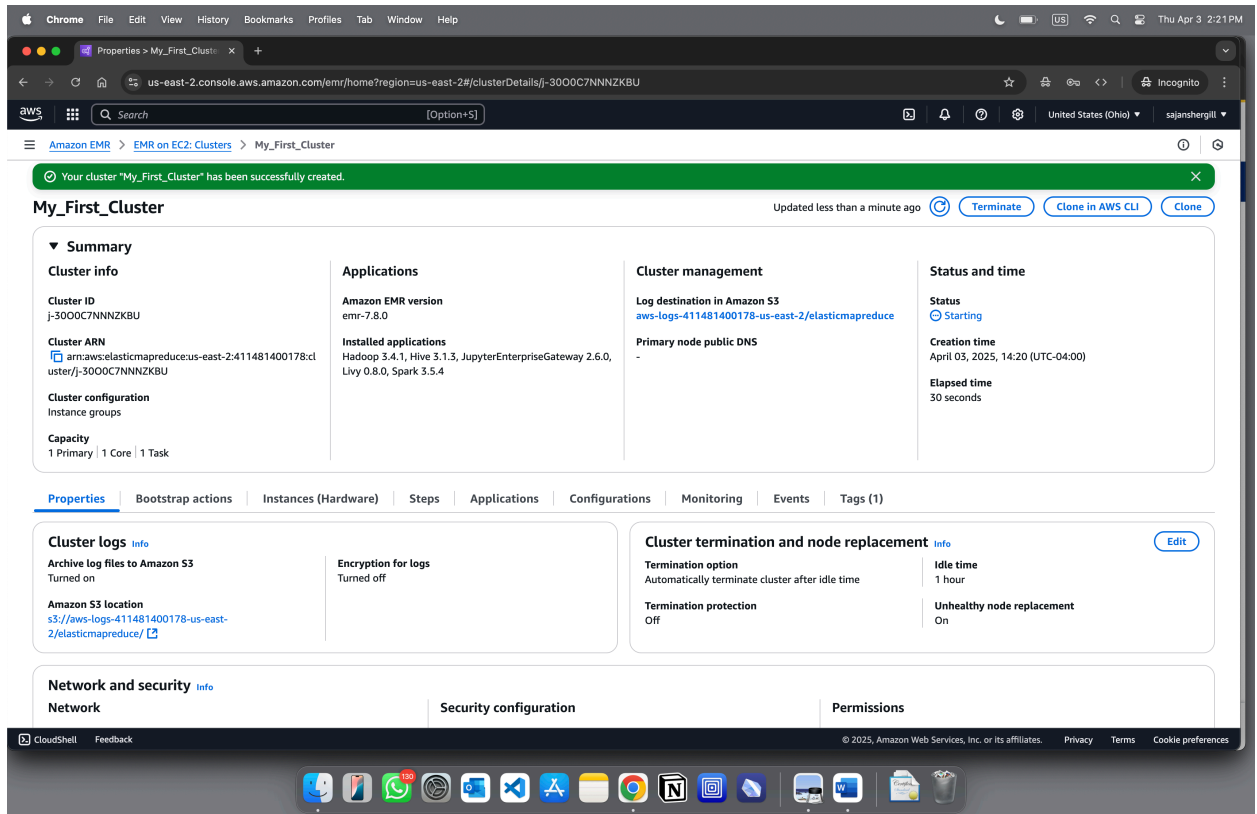
New service role

Instance profile

New instance profile

Cancel

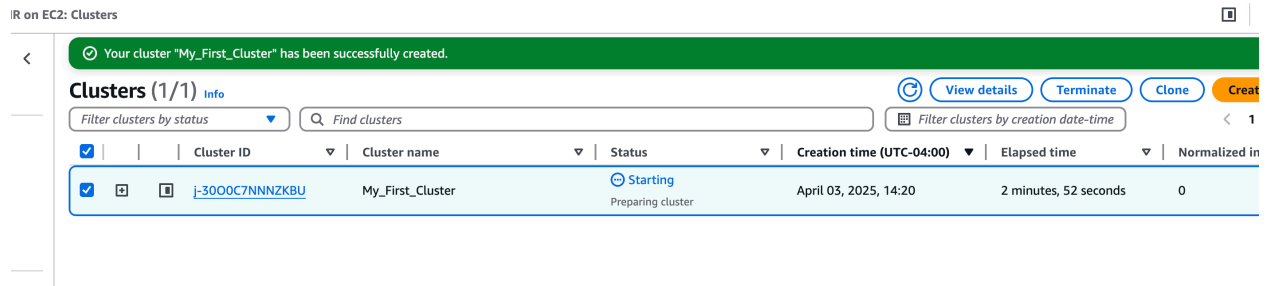
Create cluster



c. Now we need to add an inbound rule to the security group of Master Node which would allow all inbound connections on Port 22.

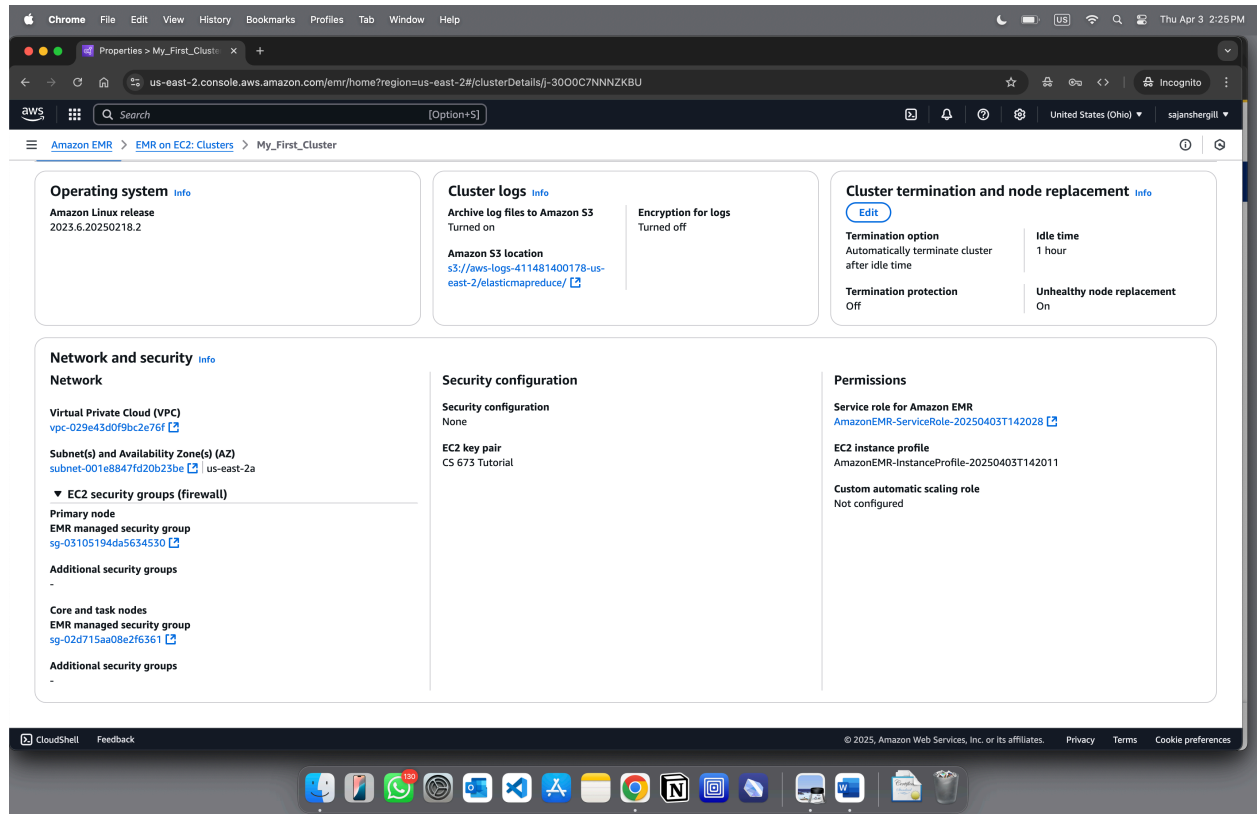
Go to EMR and click on “Clusters”

Click on the view details of the cluster which you have created.



Scroll down and check the details within “Network and security”

Click on the “Security group id” for Primary node, it will lead you to ElasticMapReduce



Scroll Down and click on “Edit inbound rules”

-> Scroll down and click on “Add rule”

-> Select Type as “SSH” and Source as “Anywhere-IPv4” as shown below:

Chrome | File | Edit | View | History | Bookmarks | Profiles | Tab | Window | Help

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#SecurityGroup:groupId=sg-03105194da5634530

EC2 > Security Groups > sg-03105194da5634530 - ElasticMapReduce-master

sg-03105194da5634530 - ElasticMapReduce-master

Details

Security group name ElasticMapReduce-master	Security group ID sg-03105194da5634530	Description Master group for Elastic MapReduce created on 2025-03-31T18:20:06.926Z	VPC ID vpc-029e43d0f9bc2e76f
Owner 411481400178	Inbound rules count 7 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (7)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
☐	-	sgr-068ee703b4efa7f82	-	All ICMP - IPv4	ICMP	All	sg-02d715aa08
☐	-	sgr-010b90226b7a3b920	-	All UDP	UDP	0 - 65535	sg-03105194da
☐	-	sgr-02d5718ce481b7d2c	-	All TCP	TCP	0 - 65535	sg-03105194da
☐	-	sgr-02583596ed0efe5af	-	All TCP	TCP	0 - 65535	sg-02d715aa08
☐	-	sgr-0cef4306d7b6a90bd	-	Custom TCP	TCP	8443	pl-eca74285
☐	-	sgr-0b7b42b0d16c4215f	-	All ICMP - IPv4	ICMP	All	sg-03105194da
☐	-	sgr-0d8c56ca921fb5741	-	All UDP	UDP	0 - 65535	sg-02d715aa08

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Chrome | File | Edit | View | History | Bookmarks | Profiles | Tab | Window | Help

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ModifyInboundSecurityGroupRules:securityGroupId=sg-03105194da5634530

EC2 > Security Groups > sg-03105194da5634530 - ElasticMapReduce-master > Edit inbound rules

sg-010b90226b7a3b920	All UDP	UDP	0 - 65535	Custom	sg-02d715aa08e2f6361		Delete
sg-02d5718ce481b7d2c	All TCP	TCP	0 - 65535	Custom	sg-03105194da5634530		Delete
sg-02583596ed0efe5af	All TCP	TCP	0 - 65535	Custom	sg-03105194da5634530		Delete
sg-0cef4306d7b6a90bd	Custom TCP	TCP	8443	Custom	sg-02d715aa08e2f6361		Delete
sg-0b7b42b0d16c4215f	All ICMP - IPv4	ICMP	All	Custom	pl-eca74285		Delete
sg-0d8c56ca921fb5741	All UDP	UDP	0 - 65535	Custom	sg-03105194da5634530		Delete
					sg-02d715aa08e2f6361		

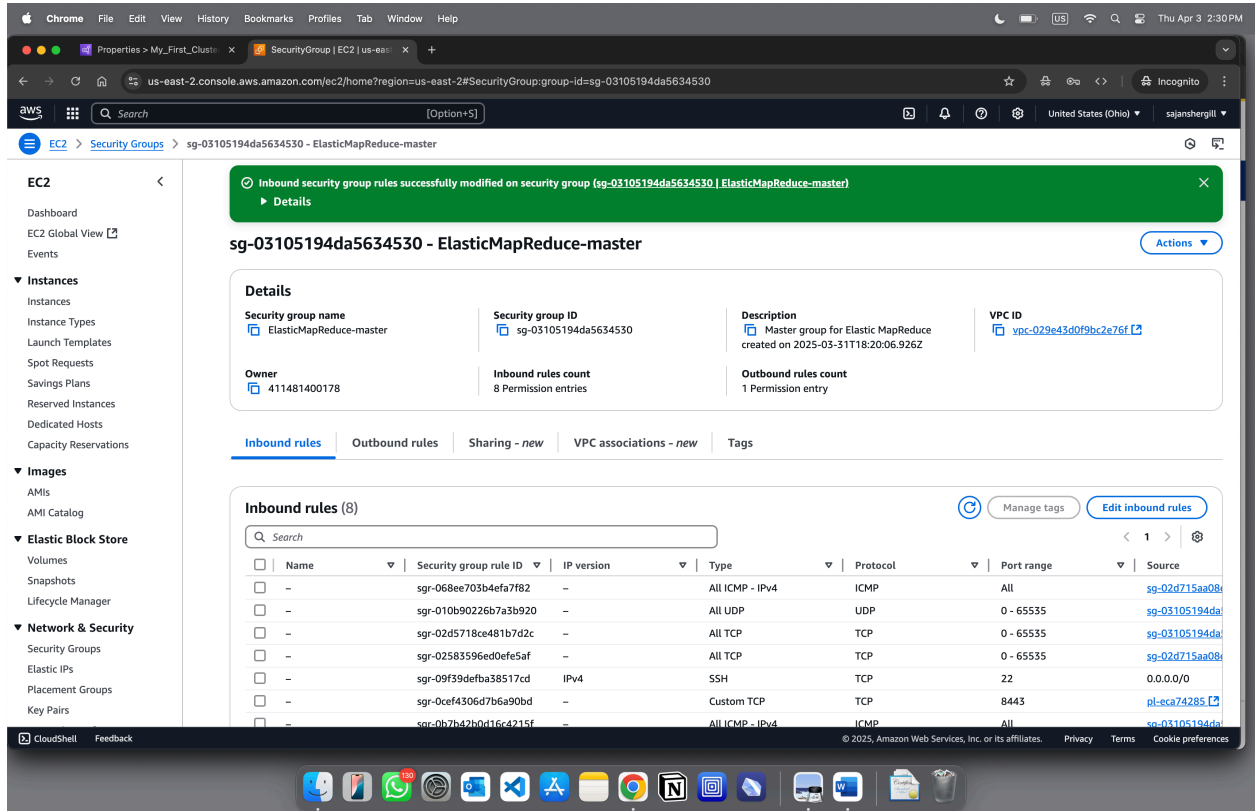
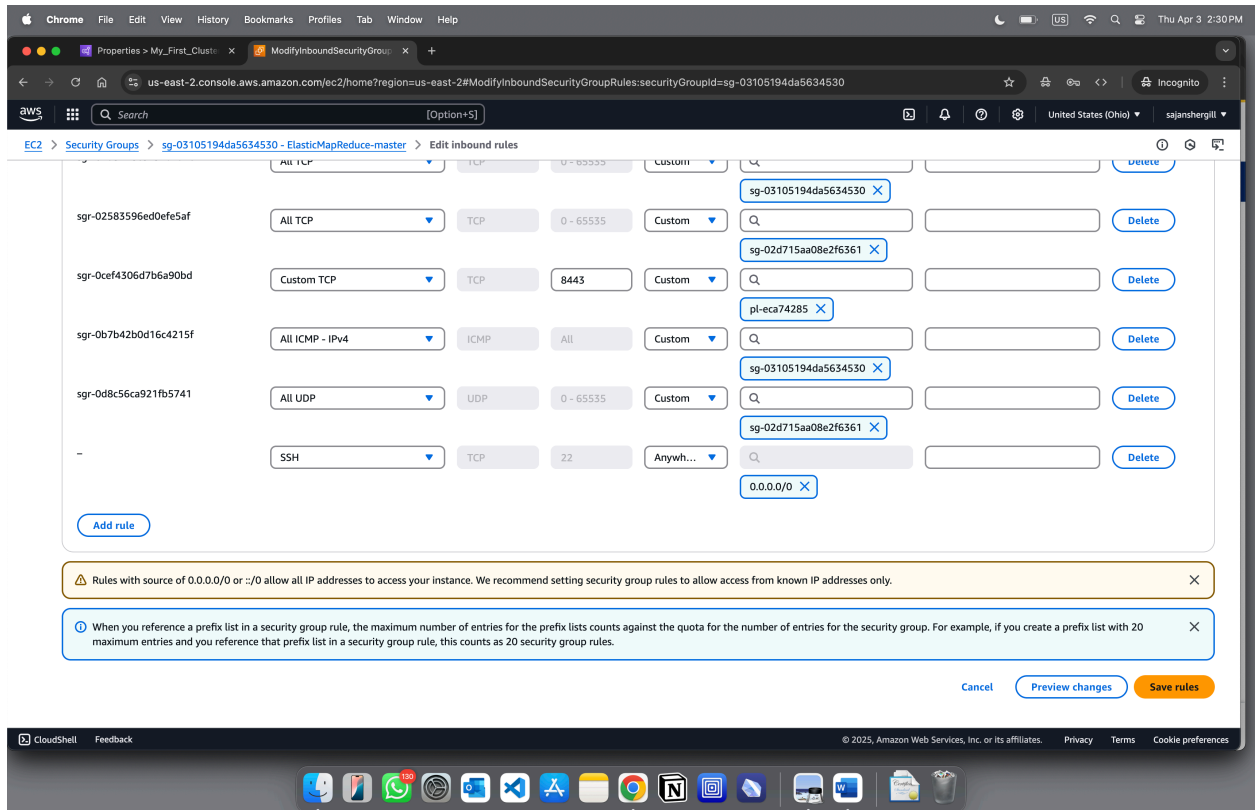
Add rule

When you reference a prefix list in a security group rule, the maximum number of entries for the prefix lists counts against the quota for the number of entries for the security group. For example, if you create a prefix list with 20 maximum entries and you reference that prefix list in a security group rule, this counts as 20 security group rules.

Cancel | Preview changes | Save rules

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Click on "Save rules"

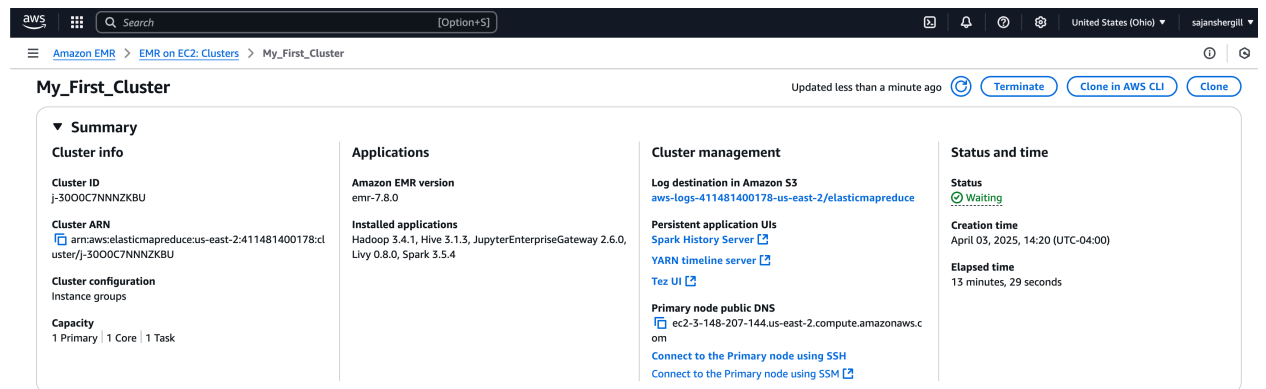
Connection to the EMR master node

3. Now connect to the EMR master node via putty to perform Hive table creation and ingestion of data

a. Search “EMR”

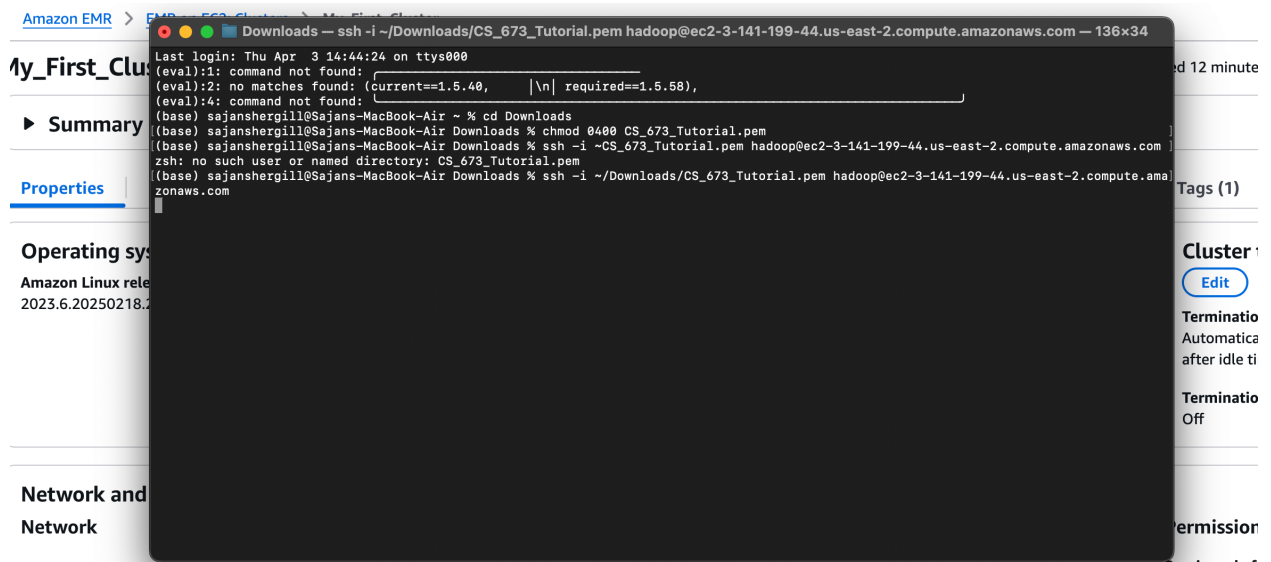
-> Click on Clusters Click on “View Cluster Details” . You will be able to see “Summary Information”

b. Click on “Connect to Master Node using SSH” , there will be your own link.



c. Open a terminal window. On Mac OS X, choose Applications > Utilities > Terminal.

d. To establish a connection to the primary node, enter the following command. Replace ~/CS673_AWS_Tutorial.pem with the location and filename of the private key file (.pem) that you used to launch the cluster.



hdfs dfs -chmod g+w /user/hive/warehouse

2. Extract the features.zip file and unzip it

a. Run the below command: wget

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/samples/features.zip>

```
[hadoop@ip-172-31-6-17 ~]$ hdfs dfs -mkdir -p /user/hive/warehouse
[hadoop@ip-172-31-6-17 ~]$ hdfs dfs -chmod g+w /user/hive/warehouse
[hadoop@ip-172-31-6-17 ~]$ wget https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/samples/features.zip
--2025-04-03 19:15:13-- https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/samples/features.zip
Resolving docs.aws.amazon.com (docs.aws.amazon.com)... 18.238.25.19, 18.238.25.28, 18.238.25.69, ...
Connecting to docs.aws.amazon.com (docs.aws.amazon.com)|18.238.25.19|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14307 (14K) [application/zip]
Saving to: 'features.zip'

features.zip           100%[=====>]  13.97K  --.-KB/s    in 0s
2025-04-03 19:15:14 (192 MB/s) - 'features.zip' saved [14307/14307]

[hadoop@ip-172-31-6-17 ~]$ unzip features.zip
Archive:  features.zip
  inflating: features.txt
[hadoop@ip-172-31-6-17 ~]$ hive
Hive Session ID = 9d1d6ab0-8e34-4d51-8504-f51605e7baaf
Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive>
```

3. Create a Hive Table and ingest data into it:

a. Type “Hive” and hit enter

b. Create a table named “hive_features” using the below DDL:

```
CREATE TABLE hive_features (feature_id BIGINT,feature_name STRING
,feature_class
```

```
STRING,state_alpha STRING, prim_lat_dec DOUBLE, prim_long_dec DOUBLE
,elev_in_ft BIGINT)
```

```
ROW FORMAT DELIMITED FIELDS TERMINATED BY '|' LINES TERMINATED BY '\n';
```

c. Load the “features.txt” data within this table

```
LOAD DATA LOCAL INPATH '
```

```
./features.txt' OVERWRITE INTO TABLE hive_features;
```

e. Run the below query to display a list of states and number of geographic features in each of these

states:

```
SELECT state_alpha, COUNT(*) FROM hive_features GROUP BY state_alpha;
```

```
hive> CREATE TABLE hive_features (feature_id BIGINT,feature_name STRING ,feature_class
> STRING,state_alpha STRING, prim_lat_dec DOUBLE, prim_long_dec DOUBLE ,elev_in_ft BIGINT)
> ROW FORMAT DELIMITED FIELDS TERMINATED BY '|' LINES TERMINATED BY '\n';
OK
Time taken: 0.415 seconds
hive>
```

```
Time taken: 0.415 seconds
hive> LOAD DATA LOCAL INPATH '
> ./features.txt' OVERWRITE INTO TABLE hive_features;
FAILED: SemanticException Line 1:23 Invalid path ''
./features.txt': No files matching path file:/home/hadoop/%0A./features.txt
hive> SELECT state_alpha, COUNT(*) FROM hive_features GROUP BY state_alpha;
Query ID = hadoop_20250403191952_b889eb5d-6ebe-4a49-ba40-650737c1c1a2
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1743707288787_0001)

-----
VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1          container  SUCCEEDED      0          0          0          0          0          0
Reducer 2 ..... container  SUCCEEDED      2          2          0          0          0          0
-----
VERTICES: 01/02 [=====>] 100% ELAPSED TIME: 6.07 s
-----
OK
Time taken: 8.944 seconds
hive>
```